COHOMOLOGY ON AN OBJECT AGREES WITH COHOMOLOGY ON ITS IMAGE UNDER A CONTINUOUS COCONTINUOUS FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the bounded below case with Leray's acyclicity lemma, the reduction of (1) to (2), and the displayed adjunction chain for the unbounded case.

Editorial note. The label lemma-pullback-same-cohomology names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Acyclicity of the pullback of an injective module. Same chapter, the lemma stating that the pullback along a continuous cocontinuous functor of an injective module has no higher cohomology on the objects of the source site. Supplies the input for Leray's acyclicity lemma in the bounded below case.
- Existence of the derived lower shriek and its adjunction. Same chapter, the lemma constructing the derived lower shriek as a left adjoint to the pullback. Supplies the middle equality of the displayed chain in the unbounded case.

Source reference. Original reference 07AC: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 9310–9310. Prerequisite pointer invoked at proof line(s) 9539; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D6X: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 9278–9278. Prerequisite pointer invoked at proof line(s) 9528; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $u : \mathcal{C} \to \mathcal{D}$ be a continuous and cocontinuous functor of sites. Let $g : Sh(\mathcal{C}) \to Sh(\mathcal{D})$ be the corresponding morphism of topoi. Let $U \in \mathrm{Ob}(\mathcal{C})$.*

(1) *For M in $D(\mathcal{D})$ we have $R\Gamma(U, g^{-1}M) = R\Gamma(u(U), M)$.*
(2) *If $\mathcal{O}_\mathcal{D}$ is a sheaf of rings and $\mathcal{O}_\mathcal{C} = g^{-1}\mathcal{O}_\mathcal{D}$, then for M in $D(\mathcal{O}_\mathcal{D})$ we have $R\Gamma(U, g^*M) = R\Gamma(u(U), M)$.*

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Proof. In the bounded below case (1) and (2) can be seen by representing K by a bounded below complex of injectives and using Lemma 0D6X as well as Leray's acyclicity lemma. In the unbounded case, first note that (1) is a special case of (2). For (2) we can use

$$R\Gamma(U, g^*M) = R\operatorname{Hom}_{\mathcal{O}_\mathcal{C}}(j_{U!}\mathcal{O}_U, g^*M) = R\operatorname{Hom}_{\mathcal{O}_\mathcal{D}}(j_{u(U)!}\mathcal{O}_{u(U)}, M) = R\Gamma(u(U), M)$$

where the middle equality is a consequence of Lemma 07AC. □